

Counting Vowels in Words

Question 2: 2025 Jan OPPE 1 - Set 1 – Section 2

Problem Statement

Write a program that processes a string and counts vowels in every word.

Input

Take a string as user input

Process

Count vowels in every word

Output

Print each word with vowel count

📌 **Important:** Words are separated by spaces and spaces must be retained in the output. This is not a function—you must write full input and output code.

Example

Input

```
Computer science is amazing
```

Output

```
Computer(3) science(3) is(1) amazing(3)
```

Key Observations

- Each word maintains its original position
- After each word, we add (vowel_count)
- Words remain separated by spaces in the output

What Are Vowels?



Lowercase Vowels

a, e, i, o, u



Uppercase Vowels

A, E, I, O, U

Vowels are the five letters that represent open vocal sounds. In programming, we must count both lowercase and uppercase versions.

Safe Approach: Create a string containing all vowels: `vowels = "aeiouAEIOU"`

Five-Step Solution Plan

01

Take Input

Get the string from user

02

Split Into Words

Break string into list

03

Count Vowels

Loop through each character

04

Format Output

Create word(count) pattern

05

Print with Spaces

Join all words with spaces

Remember: Break it down logically. Don't try to do everything in one line—clarity is more important than cleverness.

Steps 1-2: Getting Started

Step 1: Take Input

```
s = input()
```

This captures the entire string from the user.

Step 2: Split Into Words

```
words = s.split()
```

Splits the string into a list of words using spaces as delimiters.

Example Result

```
"Computer science"
```

becomes

```
["Computer", "science"]
```



The `split()` method automatically handles multiple spaces and removes extra whitespace.

Step 3: Count Vowels in a Word

The Logic

For each character in the word, check if it's a vowel. If yes, increment the counter.

```
count = 0
for ch in word:
    if ch in vowels:
        count += 1
```

Initialize Counter

Set `count = 0` before checking each word

Loop Through Characters

Examine each character in the word

Check & Count

If character is in vowels string, add 1 to count

Step 4: Print in Correct Format



Build the String

Combine word, parentheses, and count

Print with Space

Use `end=" "` to add space between words

Format Code

```
print(word + "(" + str(count) + ")", end=" ")
```

Important Notes

- Convert count to string with `str()`
- `end=" "` ensures space between words instead of newline
- No extra spaces at beginning or end

Complete Solution

Classic Approach (Exam-Safe)

```
s = input()
vowels = "aeiouAEIOU"
words = s.split()

for word in words:
    count = 0
    for ch in word:
        if ch in vowels:
            count += 1
    print(word + "(" + str(count) + ")", end=" ")
```

Why This Version?

- Simple and clear logic
- Easy to debug and understand
- Safe for exams
- Follows step-by-step structure

One-Liner Alternative

```
s = input()
for word in s.split():
    count = sum(1 for ch in word if ch in vowels)
    print(f"{word}({count})", end=" ")
```

More Pythonic but less beginner-friendly

Common Mistakes to Avoid

Missing Uppercase Vowels Forgetting to include A, E, I, O, U in vowel string	Extra Newlines Forgetting <code>end=" "</code> causes each word on new line
Wrong Loop Placement Putting <code>print</code> inside character loop prints multiple times	Entire String Count Counting vowels in whole string instead of per word
Missing Spaces Forgetting to preserve spaces between words in output	No Reset Forgetting to reset count to 0 for each new word

Logic Flow Summary

